

SAARTHI: An AI-Powered Industrial Safety Copilot for Real-Time Excavator Hazard Detection

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Abstract—Write the abstract here.

Index Terms—Industrial Safety, Computer Vision, YOLOv8, Excavator, Hazard Detection, Edge AI, Large Language Models

I. INTRODUCTION

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II. RELATED WORK

Securing heavy-machinery operations sits at the intersection of computer vision, physiological monitoring, and adaptive decision-making. Broader surveys of industrial safety still tend to focus on rule-based alarms and single-sensor approaches, and even where hybrid deep learning architectures have made real progress on driver drowsiness detection in automotive settings, they are built around a single modality. That assumption breaks down on a construction site, a mine, or a factory floor, where operators differ widely in experience and ability and hazards rarely show up one at a time. This section works through that prior literature and situates the design choices behind SAARTHI against it.

A. Threshold-Based and Infrastructure-Centric Safety Systems

Most collision-prevention systems on active worksites still come down to proximity alarms and geofencing: fixed ultrasonic or radar sensors mounted on the machine, scanning continuously and comparing whatever they detect against a static, manually set threshold, with no regard for who's actually operating the machine. In practice, those fixed thresholds tend to misfire constantly on busy sites, where workers and equipment are always moving through the sensor's range. Worse, the same threshold gets applied to every operator regardless of experience or physical ability, so an experienced operator ends up tuning out the alarms while a newer one may not get warned early enough.

Newer cabin-integrated systems have brought computer vision into the mix, using object detection to pick out workers, vehicles, and structural hazards around the machine, and drawing on signals like facial landmark drift, eyelid closure, or chassis tilt to flag unsafe states. But most of these still work in isolation. A camera watching for fatigue has no idea there's a person standing in the machine's blind spot, and a proximity sensor has no way of knowing the operator is falling asleep. Because these hazards arise from genuinely different physical processes, no single detector is well positioned to catch all of them—which is really the argument for fusing

several lightweight signals rather than trying to build one model that does everything.

Conventional alarm-based systems, built for fairly controlled, homogeneous environments, don't hold up well once the operator population and the worksite itself become this varied.

- **Rigidity and operator homogeneity.** A fixed-threshold system treats every operator identically, regardless of experience, sensory ability, or fatigue tolerance. In practice that means the same threshold that produces alarm fatigue in a veteran operator will under-warn someone newer to the job—there's no way to individualize the safeguard without changing the underlying logic entirely.
- **Deployment complexity and accessibility gaps.** Combining vision, physiological, and environmental sensing into one cabin system is already a nontrivial integration problem, and most existing architectures weren't designed with that convergence in mind. Accessibility tends to be an afterthought too: operators with hearing or visual impairments are rarely considered when alert delivery is designed almost exclusively around audio cues.

B. Fatigue-Centric and Vision-Based Defenses

A separate line of work moves detection logic onto the operator's own physiological state rather than the surrounding environment. Here the typical approach is a camera-based agent watching the driver's face—tracking eye-aspect-ratio, yawning frequency, head pose—to infer drowsiness before it becomes dangerous.

These systems are reasonably good at what they do, but that's also their limitation: they know almost nothing about what's happening outside the cabin. A fatigue detector can't tell you there's a worker nearby, or that the terrain has become unstable, or that the machine itself is tilting toward a rollover. It's built to catch one kind of hazard and stays blind to the rest.

- **Modality opacity.** Vision-only systems generally ignore machine-state telemetry altogether, even though rollover risk from chassis inclination is well documented in the tilt-sensing literature. A fatigue model has no access to that data by design, so it simply can't detect a compound hazard—drowsiness plus instability, say—even when both are present at once.
- **The context gap.** These tools are built around a single-hazard focus and stay that way in deployment. They fire only on drowsiness cues, missing nearby obstacles or

unstable ground entirely. By the time a purely fatigue-driven alert goes off, the operator may already be in a situation involving more than just tiredness.

C. Machine Learning in Industrial Safety

As static, rule-based systems have repeatedly failed to keep up with a workforce this varied in experience and accessibility needs, machine learning has become the default approach for adaptive intervention.

1) *Deep Learning and Object Detection Models*: Deep learning dominates recent work in this space. YOLO-based detectors, for instance, perform well in real time for identifying blind-spot obstacles and nearby workers. On the physiological side, convolutional facial-landmark models track eye and head movement continuously to catch fatigue before it becomes critical, and gradient-boosted models like XGBoost have proven useful for context-aware risk scoring, weighing environmental and operational variables against historical incident records.

The catch is that these models are almost always deployed as separate pipelines. Nobody is fusing their outputs into one coherent decision—which leaves a human supervisor to piece together several unrelated alerts in real time.

- **Fragmented decision-making.** Running fatigue, obstacle, and tilt detection as three independent systems means a supervisor has to mentally combine uncorrelated alerts on the fly. That adds real cognitive load, slows reaction time, and raises the odds that a compound hazard slips through unnoticed.
- **Static intervention policy.** Even where detection is sophisticated, the response usually isn't: cabins tend to issue the same alert regardless of context, which is exactly the kind of rigid logic an adaptive, constraint-respecting policy is meant to replace.

2) *Lightweight Rule-Based Models*: Some practitioners have gone the other direction, relying on fixed-weight scoring rules or manually curated alert matrices instead. These are cheap to implement and integrate easily into legacy cabin electronics, which is a real advantage.

The tradeoff is that manual thresholds don't adapt well to individual variation—different operator experience levels, different accessibility needs, changing site conditions. Keeping thresholds current across a large fleet turns into an ongoing manual calibration problem, which doesn't scale. Reinforcement learning offers a way around this, letting a system learn its own intervention policy while staying inside predefined safety bounds, though getting RL to behave safely and explainably in a live industrial setting is still an open problem—earlier work has shown that unconstrained policy learners can behave unpredictably during exploration, especially when safety constraints aren't built into training from the start.

D. The Gap Analysis

Taken together, the literature points to a fairly clear gap: nobody has really combined multiple hazard-detection modalities, adaptive decision-making, explainability, and operator-specific accessibility into one system. Most work picks one

of these and pushes it further, while the others—particularly personalization—stay largely unaddressed.

SAARTHI is built around closing that specific gap. It fuses fatigue, blind-spot, tilt, and contextual risk signals into a single Reinforcement Learning policy operating under safety constraints, giving it the kind of adaptive intervention selection that has mostly been left to human judgment after the fact, while keeping the decision process explainable enough for safety-critical use. Rather than treating accessibility as something bolted on afterward, operator-specific delivery preferences are built directly into the Hybrid Decision Engine—so the same underlying risk score can reach an operator through haptic, visual, or audio channels depending on what they actually need, without changing the safety guarantees behind the decision itself.

III. SYSTEM MODEL AND THEORETICAL FRAMEWORK

The dynamics of the driver and vehicle's surrounding environment in real-time will have to be characterized in order to mathematically model the real-world constraints of operation for traditional vehicle tracking models. Any system deployed on the edge of the car has to contend with latency and processing limitations, making it impractical to stream raw and uncompressed HD video frames or to run large and not optimized spatial temporal deep learning architectures.

The Saarthi system models the operator and machine state using localized vision primitives and synchronised telemetry, however. This technique combines all the high frequency information from the video and sensor data into a compact risk feature vector, retaining the essential hazard information including the signature of micro-sleep, blindspot obstructions and axis tilts. This section describes the hazard threat model, the theory behind real-time risk reasoning and the localised telemetry features that will be utilized to identify any imminent operational anomalies.

TABLE I
COMPARATIVE ANALYSIS OF OPERATOR SAFETY PARADIGMS

Methodology	Hardware Class	Latency	Primary Limitation
Cloud-Based Telematics	Server + Cellular Node	High	Connectivity reliance in remote sites (e.g. mines)
Wearable Biosensors	Smartwatch / EEG Cap	Low	Operator discomfort, Hygiene, Compliance drop-off
Heavy Deep Learning (CNN/ViT)	Edge (Jetson/TPU)	Medium	High computational complexity, Thermal output
Basic Machine Thresholds	Machine ECU	Low	Rigidly generic, Complete lack of behavioral insight
SAARTHI (Proposed)	Commodity CPU	Low	Requires 30s initial operator calibration

A. The Hazard Model

Threats from industrial environments and vehicles can be broadly classified into two categories based on their visibility and the computational intelligence required for reliable detection.

1) *Type 1: Overt Operational Hazards*: Type 1 hazards occur when a physical safety boundary or operational constraint is violated. Examples include a pedestrian entering a vehicle's blind spot, an operator taking both hands off the steering wheel, or any direct violation of predefined safety zones. Similar to conventional rule-based anomaly detection, these hazards exhibit clear and observable visual characteristics, making them deterministic and comparatively easier to detect.

The proposed Saarthi framework identifies such hazards in real time by comparing the object bounding boxes generated by the YOLO object detection model against predefined restricted spatial regions. Whenever an object enters a prohibited operational zone, the system immediately classifies the event as a Type 1 operational hazard and triggers an appropriate safety response.

2) *Type 2: Covert Micro-behavioral Hazards*: Type 2 hazards represent subtle behavioural or contextual anomalies that are significantly more challenging to detect. These hazards often blend with normal operating conditions and therefore cannot be identified using simple spatial rules or threshold-based computer vision techniques. For example, a driver experiencing early-stage fatigue may maintain an apparently normal posture with open eyes while gradually exhibiting behavioural patterns associated with reduced alertness. Similarly, machine vibrations or operational tilts may remain within acceptable ranges individually but collectively indicate abnormal operating conditions.

Since the visual appearance of these situations closely resembles a safe operational state, conventional computer vision methods alone are insufficient. The proposed Saarthi framework addresses this challenge by analysing the statistical characteristics and temporal evolution of driver micro-movements, including blink rate variation, gaze drift, head pose, and machine telemetry over time. This deeper behavioural analysis enables the identification of latent hazards before they evolve into critical safety incidents.

B. Anomaly Theory in Physiological and Kinematic Sense

Facial landmark coordinate features are used to model the real-time monitoring of the operator's state in a complex environment on a heavy machine. Eye Aspect Ratio (EAR) is a measure of Eye Openness at a frame. The EAR is given as:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||} \quad (1)$$

Note: p_i represents the 2D position of the eye landmarks.

If the sequence is operated by a proper and alert operator under a relatively stable sequence around an open-eye mean μ_{alert} , the blinking caused by the EAR sequence is in the natural level. This distribution is the result of the various natural physiological blinks, head pose changes, and minor changes to the lighting in the environment. So, the eye state of an alert operator is predominantly bell-shaped and symmetric, which leads to the following form of probability density function (pdf).

This framework is based on the idea that Onset Fatigue causes specific non-Gaussian distortions to the physiological data stream:

- **Microsleep Artifacts**: The prolonged eye closure is associated with severe drowsy state, exceeding the “blink window.” These result in the rapid, brief collapse of the EAR stream (known as PERCLOS) that is not created by physiological blinks that are brief and smooth.
- **Heavy-Tailed Distributions**: Such long closures result in a skewed/Heavy-Tailed eye-state distribution (with thicker lower tails). The shape of the pdf of the eye opens is altered, with the operator suddenly waking up, and trying to recalibrate to the mean open-eye EAR (μ), however, the shape is now thicker-tailed with the appearance of fatigue artifacts.

Meanwhile, **Kinematic Anomalies** (e.g., rollover risk, blind-spot intrusion) are simulated dynamically using machine telemetry and spatial bounding boxes. If the machine speed v is not zero and the tilt angle θ exceeds a safety set-point θ_{crit} , then the safety interventions are deterministic and not in accordance with the standard behavioral policies, that is, the Tier-1 safety interventions are triggered.

C. Mathematical Formalization of Features

To capture these physiological anomalies, the system extracts a feature vector $\mathbf{x}$ from a sliding temporal window $W = \{e_1, e_2, \dots, e_w\}$ of raw EAR samples. Table II defines the mathematical notation used throughout this derivation.

TABLE II
MATHEMATICAL NOTATIONS AND DEFINITIONS. THESE SYMBOLS DEFINE THE STATISTICAL MOMENTS USED TO CONSTRUCT THE FEATURE VECTOR $\mathbf{x}$, MAPPING RAW PHYSIOLOGICAL DATA TO THE DECISION SPACE.

Symbol	Definition
W	Sliding temporal window of size w
e_i	Individual EAR sample at frame i
μ	Mean EAR (First Moment)
σ	Standard Deviation (Second Moment)
S_k	Skewness (Third Standardized Moment)
K_u	Excess Kurtosis (Fourth Standardized Moment)
E_{roc}	EAR Rate of Change
$\Phi(\mathbf{x})$	Non-linear mapping function to the XGBoost feature space

1) *Skewness (S_k)*: A measure of the asymmetry of the distribution of the physiological signal. The eye openness distribution for an alert operator is usually relatively symmetric as a function of time ($S_k \approx 0$). As the operator gets tired, however, he or she becomes prone to Micro-Sleeps and long eye closures. This causes the EAR values to bias heavily towards the lower end (closed eyes), creating a non-symmetric lower tail.

$$S_k = \frac{\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^3}{\left(\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^2\right)^{3/2}} \quad (2)$$

2) *Kurtosis (K_u)*: Kurtosis serves as a primary discriminator for detecting profound drowsiness events. It measures the tailedness of the distribution.

$$K_u = \frac{\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^4}{\left(\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^2\right)^2} - 3 \quad (3)$$

The value 3 is subtracted to calculate Excess Kurtosis, normalizing the result against a standard Normal distribution. Values far removed from 0 suggest outliers that are compatible with microsleep episodes, as opposed to physiological blinking.

3) *EAR Rate of Change (E_{roc})*: The EAR Rate of Change (E_{roc}) is used to detect the dynamics of rapid physiological changes typical of fatigue transitions such as sudden jerks awake or rapid head drops:

$$|E_{roc}| = |e_t - e_{t-k}| \quad (4)$$

with $k = 5$ indicating the step stride used to capture multi-frame displacement. High values of $|E_{roc}|$ means the eye snap is rapid or that there are sudden physiological changes that are not normally observed in calm, alert operation, where the physiological state changes slowly over time.

TABLE III

SENSITIVITY ANALYSIS OF THE FEATURES. EACH FEATURE CORRESPONDS TO A PARTICULAR TYPE OF ANOMALY; MEAN EAR PROVIDES PHYSIOLOGICAL BACKGROUND INFORMATION, WHILE KURTOSIS AND SKEWNESS DETECT THE NON-GAUSSIAN FEATURES CHARACTERISTIC OF ONSET FATIGUE.

Feature	Physiological Meaning	Detection Function
Mean EAR (μ)	Baseline Openness	Establishes the mean of the EAR function to provide physiological background context.
Std. Dev. (σ)	Blink Volatility	Stability Check: High variance means that the blinking is erratic or the head pose is unstable.
Kurtosis (K_u)	Drowsiness Burstiness	Primary Discriminator: Detects profound micro-sleeps. The longer the closures, the higher the Kurtosis.
Skewness (S_k)	Distribution Symmetry	Fatigue Artifacts: Looks for asymmetry in the distribution, which identifies a closed-eye bias.
Rate of Change (E_{roc})	Physiological Velocity	Identifies rapid changes in eye movement or sudden head drops.

D. Defining “Real-Time” for Edge AI Safety

One of the first and crucial steps in Edge AI Safety is to clearly define what is meant by “Real-Time”. However, the statistical physiological analysis requires a minimum number of w frames to be able to extract a meaningful distribution to detect fatigue.

This paper defines Real-Time in the context of Operator Safety as:

$$T_{\text{inference}} \ll T_{\text{feature_window}} < T_{\text{hazard_onset}} \quad (5)$$

The data collection time ($T_{\text{feature_window}}$) is not the critical metric, but rather the Inference Latency ($T_{\text{inference}}$) is the critical metric for Real-Time in the context of Operator Safety. The continuous feature extraction window is implemented with a moving buffer. When a Critical Fatigue event occurs, or when an operator shows signs of Critical Fatigue, a verdict should be available from the Hybrid Decision Engine ($T_{\text{hazard_onset}}$). With a considerably lower algorithmic processing time for the inference (milliseconds) compared to the time it takes the human to react in order to prevent an accident (T_{human}), the system can issue a customised warning earlier. To avoid this decision logic becoming the performance obstacle, SAARTHI will reduce $T_{\text{inference}}$ by using lightweight models optimized for the edge.

IV. CONCLUSION AND FUTURE WORK

This paper introduced SAARTHI, a personalized AI safety copilot for heavy-machinery operators facing multi-hazard industrial risks. Moving away from the siloed, rule-based alarms that dominate the field, the system fuses hazard detection into a single Hybrid Decision Engine governed by a constrained Reinforcement Learning policy—an approach that tries to reconcile consistent safety enforcement with the fact that no two operators have quite the same needs.

A. Summary of Contributions

Three things stand out from the design and integration of the system.

- **Multi-modal risk fusion.** Combining fatigue, blind-spot, tilt, and contextual risk signals into one real-time safety score captures compound hazards that any single detector would miss on its own. The individual detectors supply the sensory groundwork, but it’s the fusion logic in the Hybrid Decision Engine that actually decides when intervention is warranted.
- **Balancing safety and personalization.** The PPO-based policy weighs predefined safety constraints against operator-specific delivery preferences, choosing interventions based on accessibility needs, experience level, and stated alert preferences. That’s a meaningfully different approach from the rigid, uniform alarm thresholds most systems still rely on, and it shows that transparent, explainable safety copilots are achievable without giving up consistency.
- **Scalable and accessible deployment.** Centralizing personalization in operator profiles, and surfacing decisions through a supervisor dashboard, gives SAARTHI an edge over fragmented, single-purpose safety hardware—it scales across construction, mining, and manufacturing fleets while closing an accessibility gap those single-purpose devices never addressed. The Explainable AI layer on top of that keeps both operators and supervisors able to trust why any given intervention happened.

B. Limitations and Future Work

The framework is a solid starting point, but the design process surfaced a few areas that clearly need more work.

- **Automatic operator identification.** Right now the system assumes an operator manually selects their profile at shift start—accessibility settings, preferred language, alert preferences, all of it. On a busy or rotating crew, that step gets skipped often enough that the cabin defaults to generic settings that don’t actually match whoever’s in the seat. Facial recognition for automatic operator identification is the obvious fix: detect who’s in the cabin, load their profile—including whatever language settings they’ve already configured—without requiring a manual login. It closes the gap between what the system is meant to do and what actually happens in practice.
- **Wearable sensor fusion for fatigue robustness.** The fatigue model currently leans almost entirely on facial-landmark tracking, which struggles under poor cabin

lighting, camera occlusion, or when an operator is wearing a dust mask or safety goggles. Bringing in wearable physiological sensors—heart-rate variability bands, EEG headsets—to fuse with the existing vision-based estimate should make the fatigue score noticeably more robust to exactly those failure modes.

- **Field validation and policy hardening.** So far the reinforcement learning policy has only been trained and tested in simulated, controlled cabin conditions. Real deployment across construction, mining, and manufacturing sites will expose it to dust, vibration, and lighting variability that simulation just doesn't capture well, and that gap needs to be closed before the policy can be trusted at scale. The plan is a supervised Field Calibration phase—collecting real-world incident and near-miss data to refine the XGBoost context model's risk thresholds and retrain the PPO policy under actual operating conditions, while keeping the predefined safety constraints intact throughout.